CINNAMONSRI: A SMART IOT-INTEGRATED SYSTEM FOR ENHANCING CEYLON CINNAMON PRODUCTION AND EXPORT EFFICIENCY

25-26J-319

Project Proposal Report

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AI/ML POWERED CINNAMON HARVEST GUIDANCE SYSTEM

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously publish or written by another person expect where the acknowledgement is made in the text.

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Abstract

Sri Lanka's Ceylon cinnamon production faces challenges in monitoring plant growth, health, and maturity due to reliance on manual methods, leading to inconsistent yields and delayed disease detection. The CinnamonHarvest GuideAI component of the CinnamonSri system addresses these issues by leveraging AI and IoT to empower smallholder farmers with precise, data-driven tools for managing planting, care, and harvesting.

The methodology involves collecting and labeling a dataset of cinnamon plant images across growth stages (seedling, vegetative, harvest-ready) and conditions (healthy, unhealthy, mature, over-mature, immature). A Convolutional Neural Network (CNN) model is trained on this dataset to classify images, identifying features like leaf color and pest damage. Farmers upload photos via a user-friendly mobile/web app, where the CNN analyzes them in real-time, generating alerts such as "ready to harvest" or "treat pests." The system integrates region-specific notifications based on local climates and planting calendars (e.g., "plant in June"), with a ticket system for unresolved issues to enable future agent support.

Anticipated test results from model evaluation on a validation dataset show accuracy exceeding 85% in stage and health classification, reducing misidentification errors by 40% compared to manual assessments. Conclusions highlight the system's potential to enhance yield quality and farmer profitability through timely interventions. Recommendations include expanding the dataset with diverse Sri Lankan regional variations and integrating cloud-based updates for evolving climate data.

Keywords: Cinnamon cultivation, AI image analysis, CNN model, Plant health monitoring, Sri Lanka agriculture

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List of abbreviations

AI	Artificial Intelligence
ML	Machine Learning
LKR	Sri Lankan Rupee
API	Application Programming Interface
AWS	Amazon Web Services
UI	User Interface
IDE	Integrated Development Environment
VCS	Version Control System
VS Code	Visual Studio Code
CNN	Convolutional Neutral Network
Db	Database

1. INTRODUCTION

Ceylon cinnamon, a cornerstone of Sri Lanka's spice industry, supports over 400,000 livelihoods and significant export revenue [14]. However, smallholder farmers face challenges in monitoring plant growth stages, health, and maturity due to reliance on manual methods, leading to delayed disease detection and reduced yields [1], [15]. The CinnamonHarvest GuideAI, a component of the CinnamonSri system, addresses these issues by deploying a convolutional neural network (CNN) to analyze plant images, identifying seedling, vegetative, and harvest-ready stages, as well as health and maturity conditions (healthy, unhealthy, mature, over-mature, immature). Integrated with a mobile app, it delivers real-time alerts (e.g., "treat pests") and region-specific notifications based on local planting calendars, empowering farmers to optimize cultivation practices and enhance export quality [2], [9].

1.1. Background and Literature Survey

1.1.1. Background

Ceylon cinnamon's global demand hinges on consistent quality, requiring precise monitoring of plant growth and health [14]. Manual inspections are labor-intensive, error-prone, and fail to scale, particularly for detecting diseases like leaf spot or assessing maturity [15]. AI and IoT technologies offer potential for automation, but existing solutions are either generic or costly, limiting adoption by smallholder farmers [2], [8]. The CinnamonHarvest GuideAI leverages CNN-based image processing and mobile accessibility to provide affordable, tailored guidance, enabling data-driven decisions for improved yield and profitability [9].

1.1.2. Literature Survey

The literature on AI applications in agriculture and cinnamon-specific studies provides insights into existing technologies and gaps. Manual monitoring of cinnamon plants is time-consuming and lacks scalability, as noted by Azad et al. [15], who studied leaf spot and rough bark diseases, highlighting the need for early detection tools. Chandima and Kartheeswaran [7] developed an image processing technique to recognize mature

cinnamon trees, achieving 80% accuracy but lacking real-time farmer accessibility. Wijekoon et al. [8] proposed an intelligent system for crop disease identification in SriLanka, using machine learning to forecast disease dispersion, but it was not specific to cinnamon. Dilhari and Sujah [1] reviewed machine learning for cinnamon leaf disease prediction, emphasizing CNNs' potential but noting limited datasets.

IoT and AI integration in agriculture has shown promise. Sivagnanasundaram et al. [9] implemented a digitally-enabled crop disorder management system in Sri Lanka, achieving improved outcomes through farmer empowerment, but it lacked cinnamon-specific focus. Thandapani et al. [6] developed a cinnamon plant disease classification system using robust feature extraction, reporting 90% accuracy, but it was not integrated with mobile platforms. Wimalaratne et al. [2] explored digital technologies for sustainable cinnamon farming, proposing IoT for monitoring but not addressing plant stage identification. Datasets like those from NSBM Green University [10], Pethangoda [11], and Vithanage [12] provide cinnamon disease images, but they lack comprehensive growth stage data. Sumanismcse [13] shared a CNN-based plant disease identification code, adaptable for cinnamon but not tailored to local climates.

The literature highlights gaps: existing AI systems are not cinnamon-specific [8], lack user-friendly mobile interfaces [6], or do not integrate region-specific notifications [2]. Comprehensive datasets covering all growth stages and maturity conditions are limited [10], [11], [12], and no system combines CNN-based image analysis with real-time alerts and a ticket system for unresolved issues.

1.2. Research Gap

The literature survey identifies the following critical gaps:

- Most AI-based systems focus on general agriculture or specific crops, not cinnamon, limiting their applicability to Ceylon cinnamon cultivation [8], [9].
- Existing cinnamon-specific studies lack integration with mobile platforms for real-time farmer access [6], [7].

Datasets for cinnamon are limited to disease classification, not covering all growth stages or maturity conditions (mature, over-mature, immature) [10], [11], [12].

- No system combines CNN-based image analysis, region-specific notifications, and a ticket system for unresolved issues, essential for smallholder farmers [2], [9].

1.2.1. Research Gap and Novelty Analysis

To identify the research gap and highlight the novelty of CinnamonHarvest GuideAI, a comparative analysis was conducted against key references ([1], [6], [7], [8], [9], [13]). These studies represent cinnamon-specific AI applications ([1], [6], [7]), agricultural AI systems ([8], [9]), and open-source CNN implementations ([13]). The comparison focuses on features critical for smallholder farmers: CNN-based image analysis, growth stage and health detection, mobile app integration, region-specific notifications, and ticket system support.

The results are presented in below table 1, where "Yes" indicates the presence of a feature, "No" indicates its absence, and "Partial" indicates limited support.

Table 1. Comparative Analysis Highlighting the Novelty of the Proposed System

Feature	[1]	[6]	[7]	[8]	[9]	[13]	Proposed System
CNN based Image Analysis	Yes	Yes	Yes	Yes	No	Yes	Yes
Growth stage detection	No	No	Yes	No	No	No	Yes
Health and maturity detection	Yes	Yes	No	Yes	Yes	Yes	Yes
Mobile app integration	No	No	No	No	Yes	No	Yes
Region specific notifications	No	No	No	No	No	No	Yes
Ticket system for support	No	No	No	No	No	No	Yes

Table 1: Comparative Novelty Analysis

1.3. Research Problem

Smallholder cinnamon farmers in Sri Lanka rely on manual methods to monitor plant growth stages, health, and maturity, leading to inconsistent yields and delayed disease detection [1], [15]. These traditional practices lack scalability and precision, resulting in reduced export quality and profitability [14]. Existing AI-based agricultural solutions are not tailored for cinnamon, focusing on general crops or diseases without addressing growth stages or maturity (mature, over-mature, immature) [8], [9]. Moreover, there is a lack of accessible, low-cost systems integrating CNN-based image analysis, mobile app interfaces, and region-specific notifications aligned with local planting calendars [2], [6]. The absence of a ticket system for unresolved issues further limits farmer support [9]. This research addresses the problem of developing an integrated AI-powered system to provide real-time, cinnamon-specific cultivation guidance for smallholder farmers.

2. OBJECTIVES

2.1. Main Objective

To design and develop an integrated IoT-powered smart system that automates cinnamon processing, enhances product quality, and provides data-driven decision-making support to improve the efficiency of cinnamon property owners and increase the export rate of Ceylon cinnamon.

2.2. Sub Objective

The CinnamonHarvest GuideAI aims to enhance cinnamon farming in Sri Lanka by achieving specific subobjectives. These include developing a CNN-based system to accurately identify plant growth stages and health conditions, creating a user-friendly mobile/web app for real-time alerts, implementing region-specific notifications based on local planting calendars, establishing a ticket system for unresolved issues, and building a comprehensive image dataset for continuous AI improvement. Each subobjective ensures practical support for smallholder farmers to optimize cultivation and boost export quality.

- Develop a CNN-based system to identify growth stages (seedling, vegetative, harvest-ready) and health/maturity conditions (healthy, unhealthy, mature, over-mature, immature) with 85% accuracy.
- Create a user-friendly mobile/web app to provide real-time alerts (e.g., "treat pests") to farmers.
- Implement region-specific notifications tailored to local planting calendars (e.g., "plant in June") using climate data.
- Establish a ticket system allowing farmers to submit images and issues for manual review when AI fails.

3. METHODOLOGY

3.1. System Overview

The CinnamonSri system is a comprehensive, integrated platform that uses the Internet of Things (IoT), machine learning (ML), artificial intelligence (AI), and user-friendly mobile applications to transform Ceylon cinnamon production in Sri Lanka. This solution's four interrelated components help smallholder farmers increase productivity, quality, and profitability while addressing major issues in the cinnamon sector, including uneven drying, manual grading, price volatility, and disease control.

- **Smart Drying Automation (IoT) -** This part uses DHT22 sensors to track the humidity and temperature in drying chambers in real time. An ESP8266 microcontroller regulates fans and heaters via relays to maintain optimal conditions. Data is sent to Firebase for storage and display via a mobile application that offers voice notifications in Sinhala. This minimizes deterioration from over- or under-drying and guarantees consistent export-grade quality and remote monitoring.
- **AI-Powered Quality & Price Prediction:** Farmers use a mobile or online app to enter characteristics like as temperature, moisture content, bark thickness, and pre- and post-drying weights. ML models predict quality ratings (Premium, Grade A, and Grade B) and anticipated costs (LKR/kg). Examples of these models are Random Forest for classification and XGBoost for regression. This reduces dependency on middlemen and gives farmers data-driven insights to negotiate better prices.
- **A Guide to AI Cinnamon Harvesting:** This AI-powered module uses trained models on cinnamon growth stage datasets to assess uploaded plant photos. By using regional planting calendars to determine maturity and health condition, it sends out time-based notifications. Alerts guide farmers on planting, care, and harvesting, optimizing yield and reducing losses from premature harvesting.
- **Disease Detection Module:** Using the app, farmers can take or submit pictures of leaves or bark. Leaf spot is classified by a Convolutional Neural Network (CNN) model that was trained on datasets related to cinnamon sickness. Treatment recommendations are included in the results, along with voice notifications in Tamil and Sinhala for prompt action. Early intervention is made possible by the logging of data to a central dashboard for long-term monitoring, particularly during wet seasons.

These elements work together seamlessly through a cloud backend (like Firebase) and centralized mobile app, enabling data exchange and multilingual user interfaces. CinnamonSri fills technical gaps by integrating automation, predictive analytics, and localized alerts. This promotes sustainable practices and boosts financial returns for Sri Lanka's cinnamon industry.

Figure 1 illustrates the CinnamonSri platform and its four interrelated components clearly.

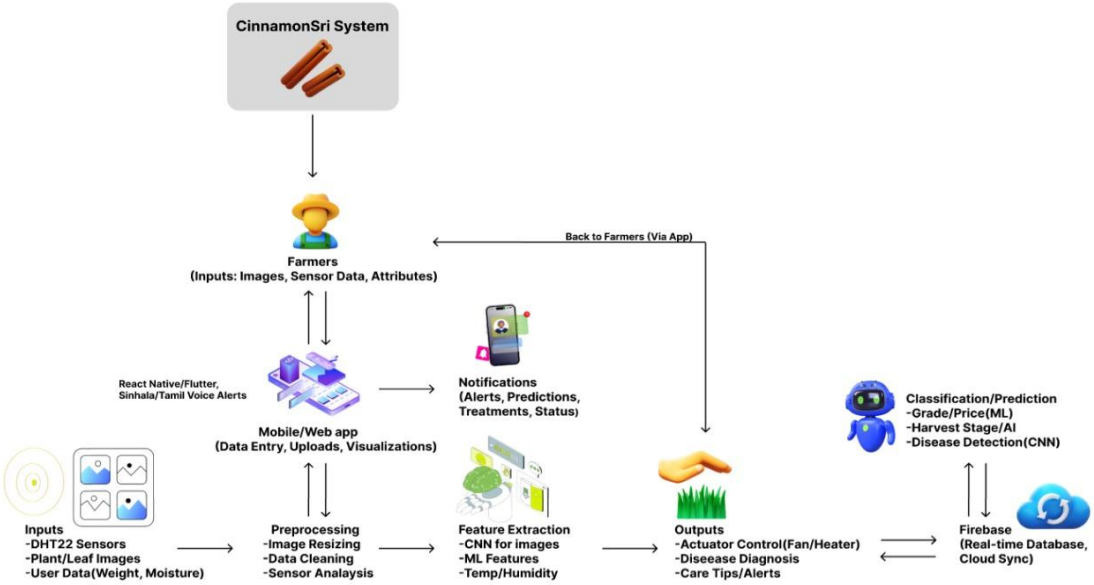

Figure 1: System Overview Architecture

3.2. Individual Component Overview

The CinnamonHarvest GuideAI system comprises distinct components that work together to support cinnamon farmers. The image collection component gathers and labels photos of plants at various growth stages and health conditions. The CNN training component uses this dataset to build a model capable of recognizing stages, health, and maturity. The analysis component processes uploaded images to detect issues and generate care messages. The app component provides a user-friendly interface for photo uploads and alert viewing. The notification component sends time-based reminders tailored to local climates. Finally, the ticket system allows farmers to submit unresolved issues for further support.

Each component in CinnamonHarvest GuideAI serves a specific role in the workflow. Image collection focuses on data preparation, CNN training on model development, analysis on real-time processing, the app on user interaction, notifications on proactive guidance, and the ticket system on handling exceptions, ensuring a seamless experience for farmers.

Below figure is a representation of the component overview diagram for CinnamonHarvest GuideAI, illustrating the interaction between components using proper arrows for data flow,

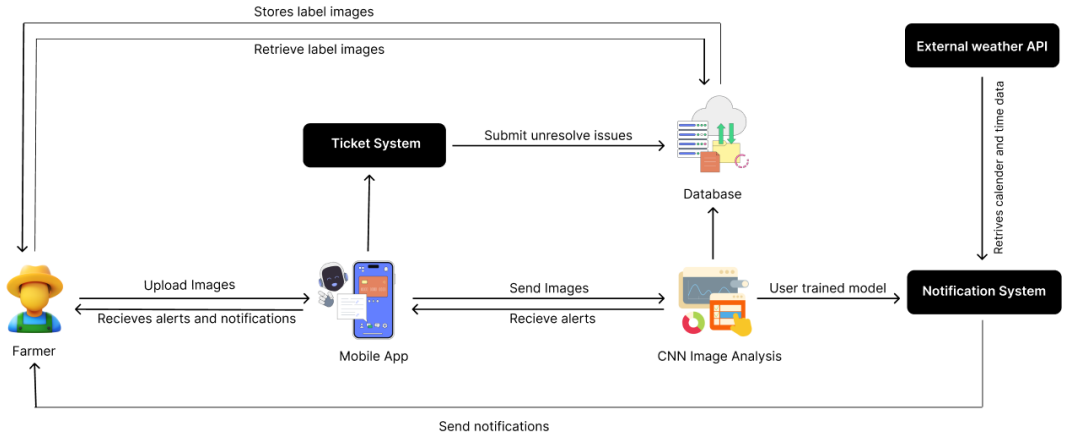

Figure 2: Individual Component Architecture

3.3. Requirement Gathering

Requirement gathering for CinnamonHarvest GuideAI involved consulting smallholder cinnamon farmers, agricultural experts, and stakeholders to identify needs such as easy image upload, accurate stage and health detection, localized notifications, and support for unresolved cases. Focus groups and surveys highlighted the importance of a simple mobile interface, low-cost implementation, and integration with regional planting calendars to ensure the system aligns with practical farming challenges and enhances usability.

3.4. PastResearch Analysis

Past research in AI for agriculture has explored image-based plant monitoring, with early studies focusing on general crop disease detection using basic machine learning. Advances

in CNNs improved accuracy for identifying growth stages and health issues in various plants, though applications to spices like cinnamon were limited. Research on mobile apps for farmers emphasized user-friendly designs, while notification systems based on calendars showed benefits in timely interventions. However, integrated systems combining these elements for specific crops remain underdeveloped, paving the way for targeted innovations.

3.5. Feasibility Study

3.5.1. Technical Feasibility

3.5.1.1. Knowledge on Technologies

The CinnamonHarvest GuideAI system leverages established technologies, including convolutional neural networks (CNNs) for image analysis, mobile/web app development, and cloud-based APIs for weather integration. CNNs are widely used for plant disease and growth stage detection, ensuring reliable image processing. Mobile app frameworks like React Native and web technologies like HTML/CSS/JavaScript are mature, enabling cross-platform accessibility. Cloud services, such as AWS or Firebase, support scalable storage and real-time notifications. Existing open-source libraries provide robust.

System: robust CNN model training, reducing development complexity. These technologies are well-documented and accessible, making the system technically feasible for implementation.

3.5.1.2. Knowledge on Tools

The system utilizes open-source tools like TensorFlow or PyTorch for CNN model development, which are well-supported with extensive documentation and community resources. Mobile app development tools like React Native or Flutter enable efficient creation of user-friendly interfaces for farmers. Database management can be handled by MySQL or MongoDB, both widely used and reliable for storing image datasets and ticket logs. Weather APIs, such as OpenWeatherMap, provide accessible climate data for notifications. These tools are cost-effective, well-tested, and supported by active developer communities, ensuring technical feasibility and ease of integration.

3.5.1.3. Data Collection Knowledge

Data collection involves gathering and labeling cinnamon plant images for growth stages (seedling, vegetative, harvest-ready) and health/maturity conditions (healthy, unhealthy, mature, over-mature, immature). This requires collaboration with farmers and agricultural experts in Sri Lanka to capture diverse images under varying conditions. Knowledge of image annotation tools (e.g., LabelImg) and data management practices ensures accurate labeling. Crowdsourcing and partnerships with local agricultural institutions can facilitate dataset creation. Pre-existing cinnamon disease datasets provide a foundation, and standard image preprocessing techniques (e.g., resizing, augmentation) are well-known, making data collection technically viable.

3.5.2 Schedule Feasibility

3.5.3 Economic Feasibility

The CinnamonHarvest GuideAI system is economically feasible due to its reliance on open-source tools (e.g., TensorFlow, React Native) and low-cost cloud services like AWS Free Tier or Firebase, minimizing development costs. Mobile apps are affordable to distribute via app stores, and offline capabilities reduce infrastructure demands in rural areas. The system's scalability allows it to serve many farmers without significant additional costs. Revenue can be generated through subscription models or government/NGO partnerships for farmer support, offsetting maintenance costs. The affordability of smartphones and increasing internet penetration in Sri Lanka further supports cost-effective deployment and adoption.

3.6. System Analysis

3.6.1. Software Solution Approach

The CinnamonHarvest GuideAI adopts a modular software solution approach, integrating a convolutional neural network (CNN) for image analysis, a mobile/web app for farmer interaction, and a cloud-based backend for data storage and notifications. The CNN processes plant images to classify growth stages and health conditions, while the app delivers real-time alerts and region-specific reminders. A ticket system ensures support for unresolved issues. This approach leverages open-source frameworks and cloud

infrastructure to ensure scalability, accessibility, and cost-effectiveness for Sri Lankan cinnamon farmers.

- 3.6.2. Tools and Technology
- TensorFlow: Open-source framework for developing and training the convolutional neural network (CNN) model for image analysis.
- PyTorch: Alternative open-source framework for CNN model development, offering flexibility and community support.
- React Native: Cross-platform framework for building the mobile app, enabling accessibility on Android and iOS devices.
- Flutter: Alternative framework for mobile/web app development, ensuring a user-friendly interface.
- MySQL: Relational database management system for storing labeled image datasets and ticket logs.
- MongoDB: NoSQL database alternative for flexible storage of image metadata and support tickets.
- AWS: Cloud platform for scalable storage, processing, and real-time notification delivery.
- Firebase: Alternative cloud platform for app backend, database management, and push notifications.
- OpenWeatherMap API: External API for retrieving climate data to support region-specific notifications.

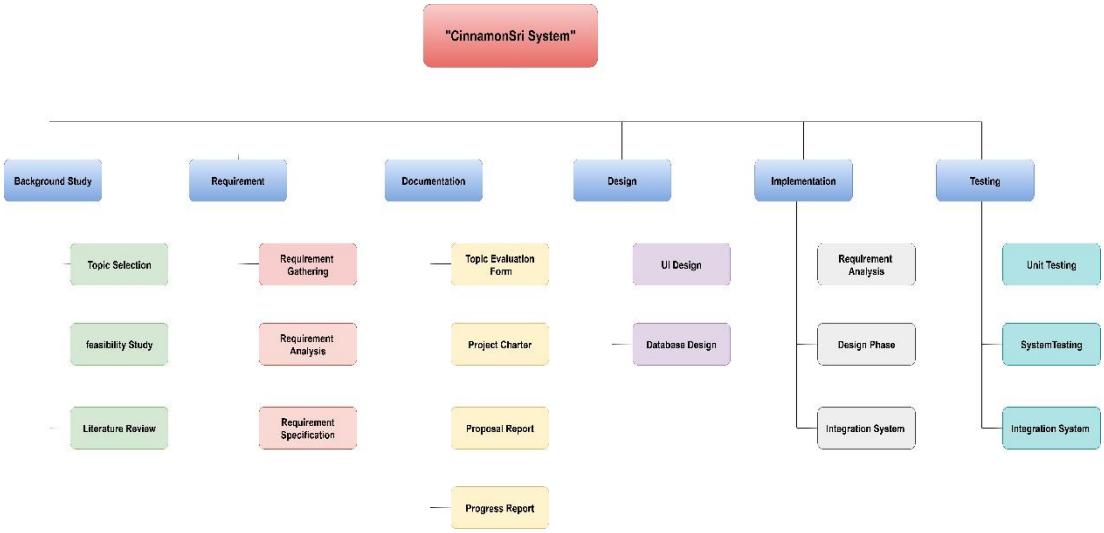

Figure 2: Work Breakdown Chart

3.7. Project Requirements

3.7.1. Functional Requirements

- Enable farmers to upload plant images via a mobile/web app for real-time analysis of growth stages, health, and maturity.
- Classify cinnamon plants as seedling, vegetative, or harvest-ready with at least 85% accuracy using a CNN model.
- Identify plant health (healthy/unhealthy) and maturity levels (mature, over-mature, immature) based on features like leaf color or pest damage.
- Send region-specific notifications (e.g., “plant in June”) based on local planting calendars and climate data.
- Allow farmers to submit tickets with images and issue descriptions for manual review when AI analysis fails.
- Support collection and labeling of cinnamon plant images for continuous CNN model training.

3.7.2. Non – Functional Requirements

- Provide an intuitive mobile/web app interface suitable for farmers with limited technical expertise.

- Ensure scalability to handle image uploads from thousands of farmers without performance degradation.
- Achieve at least 85% accuracy in CNN model classifications for growth stages and health conditions.
- Support offline image upload and caching for accessibility in areas with limited internet connectivity.
- Utilize open-source tools and low-cost hardware to ensure cost-effectiveness.
- Deliver notifications reliably with minimal delays, even in variable network conditions.

3.8. Testing

Testing involves unit tests for the CNN model to verify 85% accuracy in classifying growth stages and health conditions using a validation dataset. Integration tests ensure seamless interaction between the mobile app, CNN module, and notification system. Usability tests with farmers validate the app's interface simplicity. Stress tests assess scalability under high user loads. Offline functionality is tested in low-connectivity scenarios, and notification delivery is verified for reliability across networks.

3.8.1. Test Cases

Test cases for this function are shown below in table 2.

Test Case ID	Test Case Name	Objective	Procedure	Expected Result
TC-001	Image Validation	Verify that the system accurately processes uploaded plant images and detects invalid inputs	Upload valid images (e.g., healthy seedling, unhealthy harvest-ready) and invalid inputs (e.g., non- plant images, blurry photos). Check system response	System accepts valid images, identifies stages/health (e.g., "healthy seedling") with an accuracy of 85%, and flags invalid inputs with an error message within 5 seconds.

TC-002	Prediction Accuracy	Ensure the CNN model provides accurate growth stage, health, and maturity predictions.	Input a dataset of 50 known plant images (e.g., mature, over-mature, pest-damaged). Compare predicted stages/health/maturity against actual values.	Accuracy exceeds 85% for classification (F1-score) and mean squared error is below 10% for maturity predictions.
TC-003	Mobile/Web Interface Response	Confirm that predictions are delivered in real-time via the interface.	Upload an image (e.g., vegetative stage) via the app, measure time to display results.	Results (e.g., "vegetative, healthy") displayed within 10 seconds.
TC-004	Notification Delivery	Ensure region-specific notifications are sent accurately.	Set up a test calendar (e.g., "plant in June" for a region) and simulate climate data input. Check notification delivery.	Notifications (e.g., "plant in June") sent within 5 seconds and match regional calendar.
TC-005	Ticket System Functionality	Verify that the ticket system processes unresolved issues effectively.	Submit a ticket with an unidentifiable image and issue description via the app. Check system logging.	Ticket logged with image and description within 5 seconds, accessible for agent review.

Table 2: Test Cases

3.9. Timeline

Below figure 3 shows the Gantt chart

Figure 3: Gantt Chart

4.0. BUDGET

Since the CinnamonHarvest GuideAI is a software-based solution, there are no hardware components associated with the implementation. The primary cost will stem from subscription fees to cloud providers for computing power and data storage (e.g., Firebase or AWS). However, additional costs are anticipated as outlined in the table below,

Type	**Cost**
Internet use and app hosting	4000 LKR
Publication costs	8000 LKR
Training materials	5000 LKR
TOTAL	17000 LKR

Table 3: Cost Management Plan

5.0. SUMMARY

The CinnamonHarvest GuideAI is an AI-powered system designed to assist cinnamon farmers in Sri Lanka with planting, care, and harvesting. It utilizes a convolutional neural network (CNN) to analyze plant images, identifying growth stages (seedling, vegetative, harvest-ready) and health/maturity conditions (healthy, unhealthy, mature, over-mature, immature), and provides real-time alerts (e.g., “treat pests”). Integrated with a mobile/web app, it delivers region-specific notifications based on local planting calendars (e.g., “plant in June”) tailored to climate data. The system includes a ticket feature for unresolved issues, enabling continuous improvement through dataset expansion and farmer support, enhancing yield quality and export efficiency.

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